Information Security Education and Awareness (ISEA) Project Phase-III
(ISEA Phase-III: PMU)

Date: 30/03/2026

Template for the proposed MOOC Course and Syllabus

Important Points to remember:

i. 40 hours (20 hours video + 20 hours non-video content) / 60 hours of course content
ii. Courses in MOOCs format
iii. Guidelines of M/o Education for the development of online courses under SWAYAM must be adhered to
iv. All efforts should be made to utilize the existing infrastructure set up by M/o Education for development of MOOCs courses under NPTEL/SWAYAM initiative.
v. For online delivery of courses, necessary linkages may be established with education technology cell set up in particular institutes / nodal center coordinating SWAYAM activities.

Table-1: Course Summary

Thematic Area	**Systems and Software Security**
Institution Name & Category (Lead/Co-Lead)	Indian Institute of Technology (IIT) - Madras, Lead.
Course Title	**Functional Programming with OCaml**
Duration (No. of hours)	**60 hours of Course Content** **(30 hours video + 30 hours non-video content)**
Discipline	Secure Programming

Table-2: Course Instructor Details

Course Instructor – Please attach scanned photograph of each instructor						
S. No	**Name of the Instructor**	**Department**	**Institute**	**Mail -Id**	**Mobile number**	**Website of instructor, if any**
1	KC Sivaramakrishnan	CSE	IIT Madras	kcsrk@cse.iitm.ac.in	7305826876	https://kcsrk.info
S. No	**Name of TA**	**Department**	**Institute**	**Email id**	**Mobile number**	**Qualification**
1	Sai Venkata Krishnan V	CSE	IIT Madras	cs20d408@smail.iitm.ac.in	9499947152	PhD Student, BTech CSE
2	Vimala S	CSE	IIT Madras	cs19d750@smail.iitm.ac.in	9894247216	PhD Student

Table-3: Course Details

Target audience	● Undergraduate and postgraduate students in Computer Science and related disciplines. ● Systems engineers, software developers, and researchers. ● Professionals interested in secure and reliable software development. ● Any interested learners.
Core/Elective course	Elective
UG/PG course	UG - Core PG - Elective
Degree (s) to which course applies (BE/B. Tech/ME/M. Tech/MS/PhD etc.)	BE/BTech/ME/MTech/MS
Pre-requisites for the Course educational qualifications, prior knowledge, prior course/s, etc. if any)	C Programming, Data structures and Algorithms
Industry Linkages (List of companies / industry that will recognize/endorse the online course)	The course is relevant to organizations working in systems software, functional programming, distributed systems, and secure software development, including companies such as Microsoft, Google, Amazon, and startups in systems and cloud infrastructure domains.
Time frame for launching the course	July 2026
Whether the final certification exam be online or offline?	Offline
Weightage of assignments and final exam score to get certificate score	25% hands-on programming assignments, 75% final exam/project

Table-4: Course Outline

Modules/Durations	**Module 1: s**	**Module 2:**	**Module 3:**	**Module 4:**	**Module 5:**
Week 1	Intro to Functional Programming				
Assignments*	MCQ/quiz on functional principles				
Week 2	Expressions				

Assignments*	Coding exercises on expressions				
Week 3	Functions				
Assignments*	Function implementation tasks with test cases				
Week 4	Data types				
Assignments*	Programming tasks using lists, tuples and records				
Week 5	Pattern Matching				
Assignments*	Pattern Matching Exercises				
Week 6	Higher-order programming				
Assignments*	Hands-on programming exercise using higher-order functions in OCaml.				
Week 7		Side-effects, Modules			
Assignments*		Hands-on programming assignment using state and modules in OCaml.			
Week 8			Monads, GADTs		

Assignments*			Coding exercise on Monads/GADTs		
Week 9				Runtime, Garbage Collection, UB, Buffer Overflow	
Assignments*				Hands-on secure programming quiz / analysis exercise.	
Week 10				Memory Safety, Interfacing with C	
Assignments*				Practical assignment on C interoperability and memory-safe programming.	
Week 11				Unikernel OS	
Assignments*				Practical mini-research project on unikernels	
Week 12				Concurrency, Capability Security, DRF Safety	
Assignments*				Final hands-on project on secure concurrent programming.	

Course Outline (add/delete more weeks/modules as required depending on the course)

(*Assignments – Please fill in the assignment which would be given for that module such as MCQ, Fill blanks, Program, essay, problem solving, etc. along with evaluation mode)
NOTE: - Please extend weeks according to your course: 08 weeks for 20 hrs./12 weeks for 30 hrs.

Table-5: Course Review and Evaluation

Resources required for evaluation (TAs, software, platform, any other special requirement such as use of any software in the course) OCaml Compiler (Open-source, Linux compatible) for assignments.	
Names of 2 reviewers of the course	• Sanjiva Prasad, IIT Delhi, sanjiva@cse.iitd.ac.in • Neeldhara Misra, IIT Gandhinagar, neeldhara.m@iitgn.ac.in
Please list any special kind of question formats you may require for your course (other than MCQ, objective type answer (numeric/alphanumeric) and subjective evaluation	None
List of reference materials/books/	• **Functional Programming in OCaml**, Cornell CS3110 textbook. Freely available here. • **Real World OCaml**, by Yaron Minsky, Anil Madhavapeddy and Jason Hickey. The book is freely available at dev.realworldocaml.org.

Table-6: Course Introduction
Information for Getting the Introductory Course Page Ready for Enrollments through IVP

S. No	Description	Details
1	Course Introduction (4-5 lines)	The course teaches the principles and applications of functional programming using OCaml, with a strong focus on hands-on coding assignments, labs, and practical exercises that enable students to build secure, robust systems software. It is designed for systems engineers and developers seeking modern, security-conscious programming paradigms.
2	Two-minute introductory video about the course	
3	Course Instructor/s Photograph	
4	About the Instructor	K. C. Sivaramakrishnan is an Assistant Professor in the Department of Computer Science and Engineering at IIT Madras and the Chief Technology Officer at Tarides. He leads FP Launchpad, a centre for functional systems research and education at IIT Madras, and works on building robust, secure, and scalable systems using programming language technology.

		He led the development of Multicore OCaml, a concurrent and parallel extension of the OCaml programming language, now integrated into OCaml 5.0. His work on effect handlers has influenced modern system designs, including WebAssembly stack switching and related abstractions. Previously, he was associated with OCaml Labs at the University of Cambridge and has co-founded multiple companies, including Tarides, to advance functional systems. He holds a PhD and MS in Computer Science from Purdue University and a Bachelor of Engineering (BE) in Computer Science and Engineering from Anna University, Chennai.
5	Course plan: week wise (names of the topics)	Week 1 -- Intro to Functional Programming Week 2 -- Expressions Week 3 -- Functions Week 4 -- Data types Week 5 -- Pattern Matching Week 6 -- Higher-order programming Week 7 -- Side-effects, Modules Week 8 -- Monads, GADTs Week 9 -- Secure Programming - Runtime, Garbage collection, undefined behaviour, buffer overflow Week 10 -- Secure Programming - Memory Safety, Interfacing with C Week 11 -- Secure programming - Unikernel OS Week 12 -- Secure programming - concurrency, capability-based security, DRF freedom and safety.
6	Two sets of question papers with answer key and solution, once the course is approved by respective institutions.	

K.C. Sivaramakrishnan
Name, signature of the Faculty/Instructor/s
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